

GC University Lahore

BS Computer Science Examination: Final 2024

Semester: 5th

Course Code: CS-2205

Title: Operating System

Time Allowed: 155 minutes

Max. Marks: 40

Subjective Type

Attempt any Four Questions. All questions carry equal marks (10 Marks each).

Question No. 02 [2 + 2 + 2 + 2 + 2 = 10 Marks]

- A. What is the purpose of a system call? (2)
- B. Differentiate between a co-operating process and an independent process. (2)
- C. What are the necessary conditions for a deadlock to occur? (2)
- D. Differentiate between short-term and long-term scheduling. (2)
- E. What is a Process Control Block (PCB)? (2)

Question No. 03 [10 Marks]

Draw a Gantt chart and calculate the average waiting time for the following processes by using **FCFS**, **SJF** (Preemptive and non-preemptive), **SRTF**, and **Round Robin** (with time quantum = 2) algorithms:

Process Name	Arrival Time	Burst Time
P ₀	1	7
P ₁	2	3
P ₂	3	5
P ₃	5	1

Question No. 04 [3 + 3 + 4 = 10 Marks]

- A. What is the difference between physical and logical address space? (3)
- B. What is internal and external fragmentation? (3)
- C. Define memory selection, and elaborate on the **best-fit**, **worst-fit**, and **first-fit** algorithms. (4)

Question No. 05 [10 Marks]

Consider the following resource-allocation graph specifications:

- $P = \{P_1, P_2, P_3, P_4\};$
- $R = \{R_1, R_2, R_3\};$
- $E = \{R_1 \rightarrow P_1, P_1 \rightarrow R_2, R_2 \rightarrow P_2, P_2 \rightarrow R_3, R_3 \rightarrow P_3, P_3 \rightarrow R_1, R_1 \rightarrow P_4\};$
- Resource type R_1 has two instances
- Resource type R_2 has one instance
- Resource type R_3 has one instance

Draw the resource allocation graph and explain in detail the possibility of a deadlock.

Question No. 06 [3 + 3 + 2 + 2 = 10 Marks]

The VAX/VMS system uses a **FIFO** page replacement algorithm for resident pages and a **free-frame pool** of recently used pages. Assume that the free-frame pool is managed using the **Least Recently Used (LRU)** replacement policy.

- a) If a page fault occurs and if the page does not exist in the free-frame pool, how is free space generated for the newly requested page? (3)
- b) If a page fault occurs and if the page exists in the free-frame pool, how is the resident page set and the free-frame pool managed to make space for the requested page? (3)
- c) What does the system degenerate to if the number of resident pages is set to one? (2)
- d) What does the system degenerate to if the number of pages in the free-frame pool is zero? (2)

